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Rapport de Thèse

By: Darin A. Croft, Ph.D.
Professor and Chair, Department of Anatomy
Case Western Reserve University School of Medicine
10900 Euclid Ave., Cleveland, Ohio, USA 44106-4930

« Perspectives macro-éco- évolutives sur l'histoire des mammifères sud-américains
au cours du Cénozoïque »

Monsieur Lucas BUFFAN

Doctorat en EERGP - Écologie, Evolution, Ressources Génétique, Paléobiologie

ISEM - Institut des Sciences de l'Evolution -Montpellier
Ecole Doctorale GAIA - Biodiversité, Agriculture, Alimentation, Environnement, Terre, Eau.

This doctoral thesis by Lucas Buffan is an excellent piece of scholarship that marshals large datasets and simulations to explore several topics of broad interest to not only paleontologists but also biologists. It focuses on the fossil record of South America and spans much of the Cenozoic. The three chapters of original research (Chapters 1-3/B-D) each address a distinct temporal interval and macroevolutionary topic. They are structured as self-contained studies but are united in their macroevolutionary perspective and quantitative approaches. They complement one another quite well, resulting in a highly coherent thesis. The conclusions are based on a vast amount of cleaned and vetted data - unlike many studies, which simply assume that any data download from online databases are accurate - and a variety of analytical approaches. The introductory chapter (A) provides the reader with a thorough yet accessible review of the major topics discussed by the research chapters. It spans philosophical underpinnings, methodological considerations, and the fossil record itself. The summary chapter (E) thoughtfully synthesizes the findings of the research chapters and candidly discusses shortcomings of the studies and opportunities for future improvements.

Specific strengths:

Chapter 1 tests whether South American mammals experienced a mass extinction event at the Eocene-Oligocene Transition (EOT) in South America, as has been described for mammals on Northern Hemisphere continents. It also explores whether significant changes in trophic structure occurred during this interval and whether different patterns characterized tropical versus extratropical faunas. Although various researchers have long recognized that a "remodeling" of faunas took place during this interval, no one had previously quantified or characterized this phenomenon in terms of pattern and timing. The results obtained clarify several aspects of this transition: there was no mass extinction event at the EOT, there were no major ecological (trophic) changes during this interval (despite significant taxonomic turnover), and mammals differed in their response depending on their latitude. The novelty and importance of these results is evidenced by the fact that the chapter was published in the Proceedings of the National Academy of Sciences of the USA. The high-quality data on which these analyses were based - due to careful cleaning and vetting - will pay many dividends for science by enabling other researchers to undertake additional analyses of the "Bisagra Patagónica".

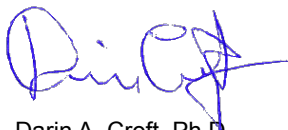
Chapter 2 analyzes diversity trends in chinchilloid rodents from the latest Eocene to the Recent, incorporating two key ecological attributes: body size and tooth height. One remarkable aspect of this clade is that extremely large size (far exceeding that of the extant capybara) evolved at least twice within the group: once among neoepiblemids and once among dinomyids. One goal of the chapter was to analyze possible factors in the extinction of these massive rodents. Two different but complementary methodological approaches were used in order to test some of the underlying assumptions: a model-based one (PyRate) and one based on biodiversity simulations (DeepDive). Although the reason(s) these giant rodents went extinct remains an open question - as is true for most extinctions in the fossil record - this study yielded a clearer picture of the context of the extinctions, finding that Andean uplift and sea-level changes were likely contributing factors. Beyond the new insights into the evolution of chinchilloids provided by this chapter, I very much appreciated the explicit demonstration and discussion of how using outdated (versus updated) taxonomy can affect resulting diversity trend. In this respect, dinomyids are an egregious example of over-splitting, but they are not the only group of South American mammals for which this is true (e.g., notoungulates). The comprehensive body mass dataset compiled for this chapter will be extremely useful for researchers interested in the paleobiology of chinchilloids and the structure of ancient mammal communities.

Chapter 3 tackles one of the most publicized biogeographic events of the Cenozoic: the Great American Biotic Interchange (GABI). Many pages have been published on this phenomenon, yet this chapter took a completely novel approach to understanding this phenomenon by comparing the results of simulations to the observed pattern of the exchange to test scenarios and explanatory hypothesis. A surprising result is that dispersals from south to north are favored over those in the opposite direction, at least when only contained by climate; this is exactly opposite of the observed pattern. This suggests that biological causes played a greater role in shaping the modern mammal faunas of both continents than climatic ones, though geographic contingency likely played a role as well. The new data and interpretations regarding the GABI in this chapter not only provide new clues about its causes, they will generate also considerable discussion and future avenues of research.

Taken as a whole, this thesis illustrates an impressive depth and breadth of knowledge of the fossil record of South American mammals and methods for analyzing fossil occurrence data. In addition, it is very clearly written and organized, with many well-designed figures and much supporting information. In my judgement, it is commensurate with the expectations of the qualifications of a doctoral candidate. Therefore, **I am strongly in favor of authorizing the oral defense of Lucas Buffan** as scheduled on Sept. 11, 2026

On the following pages, I include some questions that would be interesting to consider and perhaps incorporate into a final version of the thesis. Most of these are not required modifications; only the few errors noted should be corrected. However, I do suggest homogenizing the numbering of figures in the chapters and in supplementary information so that each is unique. For example, there is a Figure C.1 on p. 58 (in Chapter 2) as well as a Figure C1 on p. 251 (associated with Chapter 3). This is confusing and means that the only way to find the correct figure is to use the hot link in the PDF.

Sincerely,



Darin A. Croft, Ph.D.
Professor and Chair

Questions and Comments

What makes something a tropical fauna? Or, what are the tropics? Does it just come down to temperature? I appreciated the explicit and quantitative way that paleogeography and paleoclimate were used to distinguish tropical faunas from extratropical faunas in Chapter 1. What are the pros and cons of incorporating paleoclimate into this definition? Based on Figure A.3, tropical faunas are restricted to a narrow latitudinal band about 10° N/S of the Equator, far smaller than the area most geographers regard as tropical. The tropics of Cancer and Capricorn are based on celestial relationships (i.e., they represent the northernmost and southernmost extent of the sun directly overhead during the year). It seems like this is a well-established definition. Why change it?

Related to the above, what is the justification for treating all faunas outside of the tropics as the same? This latitudinal span in South America is twice that of the tropics (ca. 10-50° S). Why not subdivide it into two latitudinal bands of similar extent?

Marcot et al. (2016) found that the latitudinal diversity gradient (LDG) didn't appear until the Late Cenozoic. What evidence do we have that it existed in South America during the middle Cenozoic? Are the differences that were found simply due to chance and the relatively small sample size of tropical localities? Marcot et al. (2016) should be cited and discussed in the context of the LDG (www.pnas.org/cgi/doi/10.1073/pnas.1524750113).

How much of the observed pattern in Chapter 1 is simply due to the fossil record? This interval has many gaps of uncertain duration during the Eocene, plus a very long Desadan SALMA, which could be obscuring patterns in the latter half of the Oligocene.

On p. 41, I don't understand this statement: "According to our findings, this turnover is particularly striking at the family level, with the synchronous extinction and origination of three (Archaeopithecidae, Notostylopidae, and Oldfieldthomasiidae) and four families (Homalodotheriidae, Hegetotheriidae, Mesotheriidae, and Toxodontidae)." These do not seem to be synchronous; in the figures cited (A.19-20), the last occurrence of Archaeopithecidae is prior to the late Eocene (long before the EOT), the last occurrence of Oldfieldthomasiidae is at the EOT, and the last occurrence of Notostylopidae is in the Early Oligocene, well after the EOT. Additionally, Figs. A.19-20 look exactly the same to me, even though their captions suggest they are different. Am I missing something, or is this a mistake? (Note also the spelling of Homalodotheriidae - without the n.)

Can something be an open habitat without being a grassland? These terms seem to be used synonymously, but they are different. The former is a structural term (i.e., it refers to the structure of the vegetation), whereas the latter is a taxonomic term. Various lines of evidence indicate that Cenozoic South American floras may have had non-analog aspects compared to modern floras/biomes, so care must be taken when using these terms (cf. Strömberg et al. 2024; <https://doi.org/10.1016/j.palaeo.2024.112518>).

On p. 56, it states that "*Lagostomus* and *Dinomys* inhabit the Argentinian and the West-Amazonian plains, respectively." This isn't quite accurate; the range of *Lagostomus* extends into Bolivia and Paraguay and includes the Monte and Chaco regions in addition to the Pampas (see Jackson et al. 1996, Mammalian Species No. 543).

In chapter 2, I prefer the terms hypselodonty to euhypsodonty. See Janis and Fortelius (1988; Biological Reviews 63:197-230) for a justification for avoiding the latter term. But I realize this is only a preference.

What does the paucity of Early Miocene chinchilloids (p. 64) tell us about their paleobiology (habits and/or distribution)? This is one of the best-sampled intervals in the South American mammal fossil record, at least in some respects, so the lack of a record is surprising.

With regard to the discussion of the evolution and persistence of hypsodonty on p. 74, it seems that hypselodonty in notoungulates allowed a broadening of dietary niche (we discuss this in Croft et al. 2020). This may be due to the “fact” (as far as I am aware) that once it evolves, it is never lost. This means that the factors that favored the initial evolution of hypsodonty in the Paleogene (open habitats) may not be correlate with the presence or persistence of hypselodonty in the Neogene. This is shown by some of the studies cited (e.g., Armella and Croft 2026 did not interpret pachyrukhines as open habitat grazers).

Related to the above: Can dust or grit account for the differential evolution of hypsodonty in closed habitats (i.e., some rodents or notoungulates evolving it while others do not)? I agree that hypsodonty does not necessarily correlate with grass consumption, but I’m not aware of any instances where it has evolved in the absence of at least partially open habitats - with the possible exception of fossorial taxa.

In Chapter 3, are low and high latitude general terms, or is there some cutoff between the two? This was not entirely clear to me. I suspect they are not, but given how they are distinguished and defined in Chapter 1, this would be worth clarifying.

Related to the above, note that the tropics in this chapter (most of the area between 20° N and 20° S; see Fig. D.4A) differ significantly from the tropics in Chapter 1. Does it make sense to have different definitions for different scenarios?